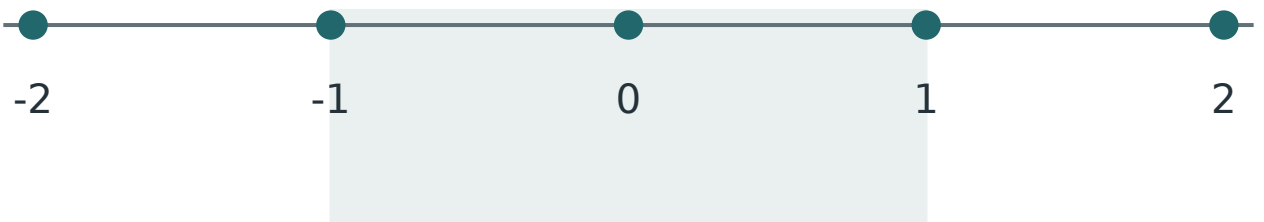
$$G = a/b \quad \Longrightarrow \quad b(AG - B) = aA - bB \in \mathbb{Z}$$

$$0 < |AG - B| < 1/b$$

would place a nonzero integer in this gap